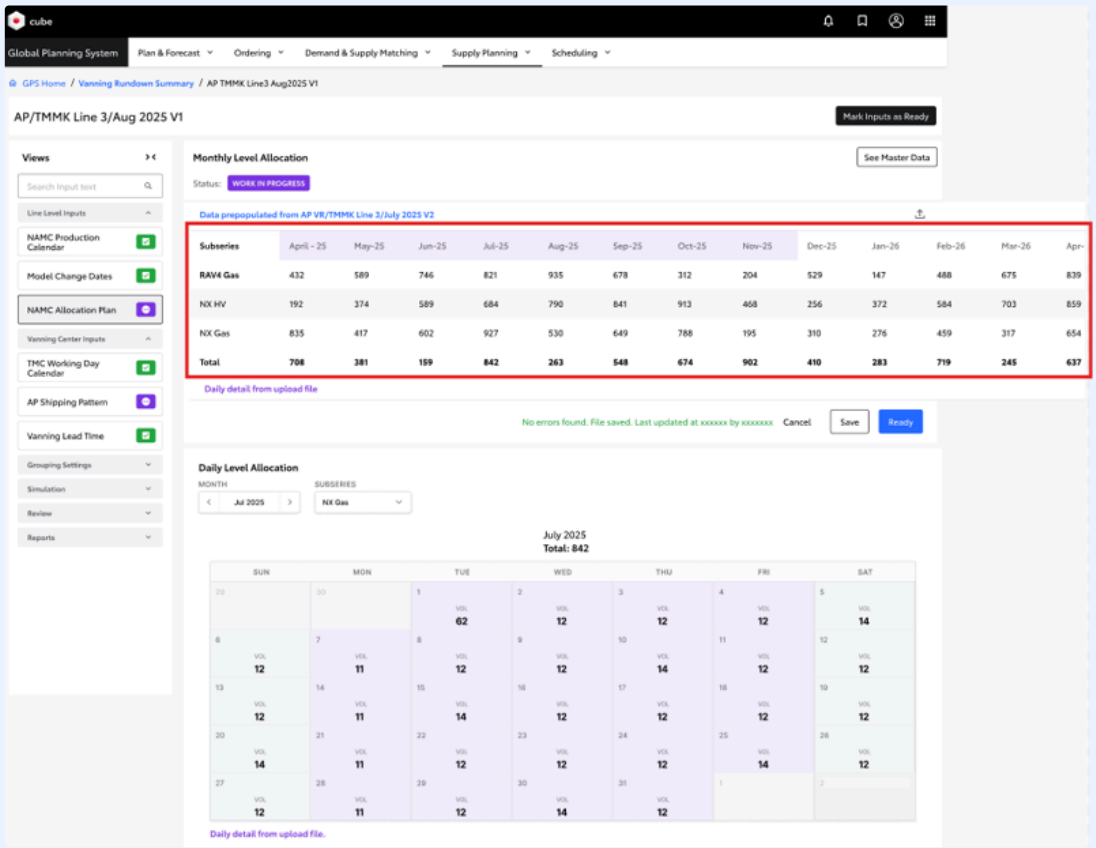
SPPL-166: NAMC Allocation Plan Get API technical workflow

Jira Ticket: SPPL-166: BE | Create technical workflow for NAMC Allocation Plan Get API on parameter inputs page including DB details IN PROGRESS

This page provides comprehensive information about the GET API for NAMC Allocation Plan on the Input params page, to view NAMC allocation plan at monthly level.



Supply Planning GET NAMC Allocation Plan API Details:

1. **Hosted URL**: https://api.spln.cube<env>.toyota.com/<env>
2. **API Name:** <HOSTED_URL>/v/1/1/sp/monthly-namc-allocation-plan
3. **Path Param**: N/A
4. **Query Param**: ?scenarioId=uniqueScenarioId&userEmail=userEmail
5. **Method:** GET
6. **Authorization**: Bearer token (**required**)
7. **Lambda Name**: spln-<env>-view-monthly-namc-allocation-plan-v1-1
8. **API Request Body**: N/A
9. **API Response:**

- **Success response with status 200:**

```json
{
  "scenarioId": "uniqueScenarioId",
  "data": [
    {
      "monthYear": "Jan-25",
      "totalMonthlyVolume": 350,
      "isEditable": true
      "subSeries": [
        {
          "name": "RAV4 Gas",
          "monthlyVolume": 200,
        },
        {
          "name": "RAV4 HV",
          "monthlyVolume": 150,
        }
      ]
    },
    {
      "monthYear": "Feb-25",
      "totalMonthlyVolume": 500,
      "isEditable": false
      "subSeries": [
        {
          "name": "RAV4 Gas",
          "monthlyVolume": 300,
        },
        {
          "name": "RAV4 HV",
          "monthlyVolume": 200,
        }
      ]
    }
  ],
  "scenarioSteps": {
    "Line Level Input": {
      "NAMC Production Calendar": "COMPLETED",
      "Model Change Dates": "COMPLETED",
      "NAMC Allocation Plan": "COMPLETED"
    },
    "Vanning Center Inputs": {
      "Shipping Pattern": "COMPLETED",
      "Vanning Lead Time": "COMPLETED",
      "TMC Working Day Calendar": "COMPLETED"
    },
    "Grouping Settings": {
      "Grouping": "COMPLETED",
      "Min Max Stock": "IN PROGRESS",
      "Fluctuation Allowance": "NOT STARTED"
    },
    "other": ["Simulation", "Review", "Reports"]
  }
}
```

- **Error Response:**
  - **any internal server error occurred with status 500:**

```json
{
  "errorMessage": "Internal Server Error"
}
```

- **any unauthorized error occurred with status 401 (Managed by the AWS API Gateway integrated Custom Authorizer):**

```
{
  "message": "Unauthorized"
}
```

- **If request is in-valid with status 400:**

```
{
  "errorMessage": [<"ValidationError: validation error message”>]
}
```

- **List of possible validation error message:**
  1. "ValidationError: scenarioId is required and must be a uuid."
  2. "ValidationError: Scenario doesn't exist."
  3. “ValidationError: userEmail is required and must be a string.”
  4. “ValidationError: Invalid userEmail.”

10. **Table:**
    a. **scenarios**
    b. **namc_allocation_plan**
    c. **scenario_step_status**

SP Fetch Monthly NAMC Allocation Plan API Workflow:

- **Technical workflow:**
  - Description:
    i. The Supply Planning user logs in to the application.
    ii. After the user provides creates a scenario and opens it, a request is sent to the SP Get NAMC Allocation Plan endpoint: ***sp/namc-allocation-plan*** with parameters: scenarioId=uniqueScenarioId&userEmail=userEmail
    iii. After successful authentication, the **spln-view-monthly-namc-allocation-plan-v1-1** lambda is triggered by the SP view scenario NAMC Production Calendar endpoint: ***sp/*monthly-*namc*-allocation-plan**. If authentication fails, an error with status 401 (*refer: error response-**any unauthorized error occurred with status 401***) will be returned and sent to client.
    iv. After user authentication has been confirmed, the **spln-view-monthly-namc-allocation-plan-v1-1** lambda will perform validation on the request payload. If payload is valid will proceed to next step else throw validation error (*refer: error response**: Section if request is in-valid with status 400 and List of possible***

validation error message). Below query is used to check if provided scenario is valid:

```
select * from supply_planning.scenarios where scenario_id = :scenarioId and
is_active = true;
```

v. After the request payload is successfully validated, the **spln-view-namc-allocation-plan-v1-1** lambda execute below queries to get scenario NAMC Production Calendar data from the **namc_allocation_plan** table

```
select * from supply_planning.namc_allocation_plan where scenario_id =
:scenarioId
and is_active = true;
```

vi. Below query is used to fetch view scenario menu data: ref (SPLN-162: View Scenario Menu technical workflow document)

```
select * from supply_planning.scenario_step_status where scenario_id =
:scenarioId;
```

vii. The retrieved data is formatted as shown below before returning the response to the client with a status code of 200:

1. If the Monthly NAMC Allocation Plan data for scenario is available:

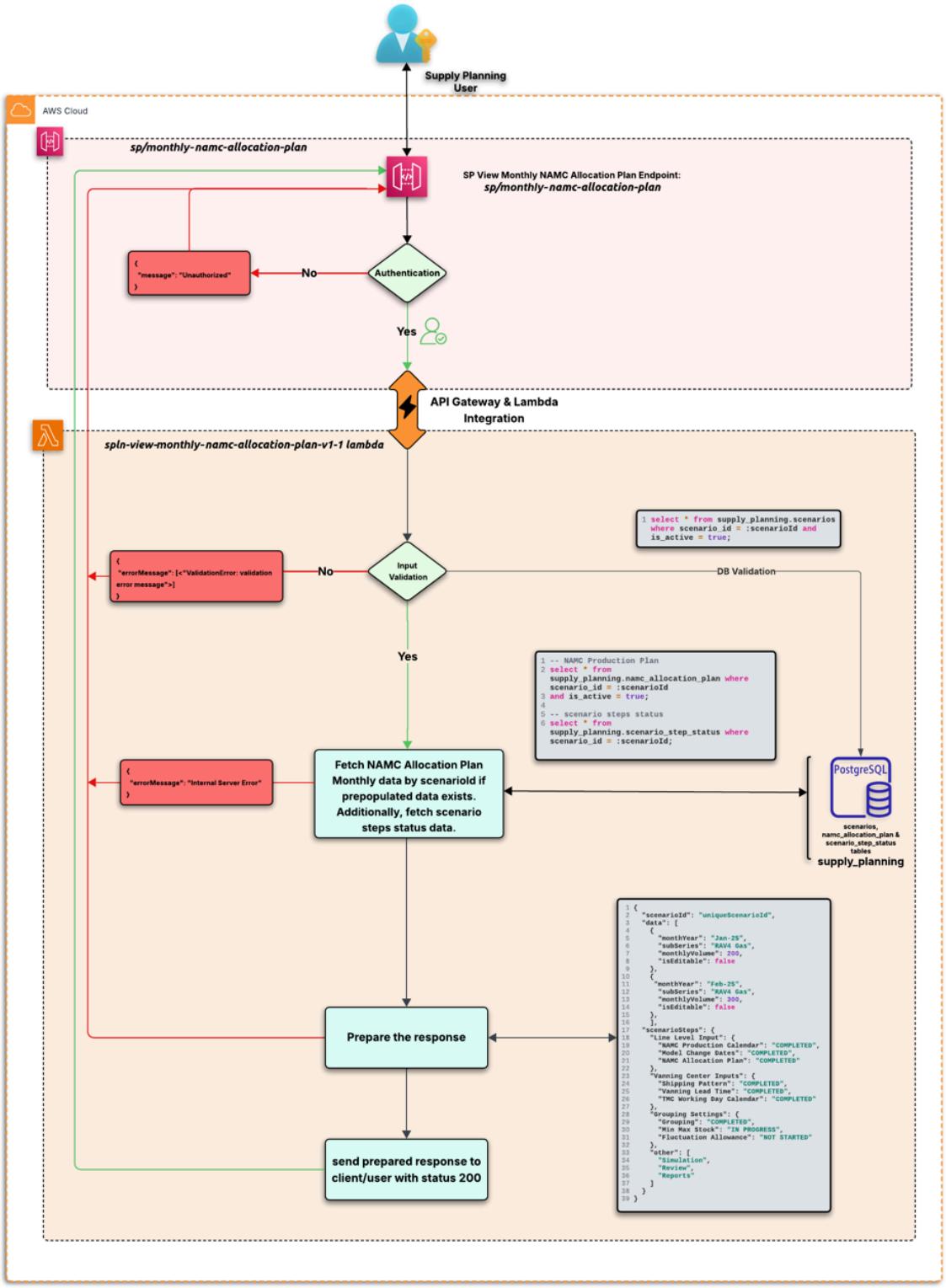
```
{
  "scenarioId": "uniqueScenarioId",
  "data": [
    {
      "monthYear": "Jan-25",
      "totalMonthlyVolume": 350,
      "isEditable": true
      "subSeries": [
        {
          "name": "RAV4 Gas",
          "monthlyVolume": 200,
        },
        {
          "name": "RAV4 HV",
          "monthlyVolume": 150,
        }
      ]
    },
    {
      "monthYear": "Feb-25",
      "totalMonthlyVolume": 500,
      "isEditable": false
      "subSeries": [
        {
          "name": "RAV4 Gas",
          "monthlyVolume": 300,
        },
        {
          "name": "RAV4 HV",
          "monthlyVolume": 200,
        }
      ]
    }
  ],
  "scenarioSteps": {
```

```
    "Line Level Input": {
      "NAMC Production Calendar": "COMPLETED",
      "Model Change Dates": "COMPLETED",
      "NAMC Allocation Plan": "COMPLETED"
    },
    "Vanning Center Inputs": {
      "Shipping Pattern": "COMPLETED",
      "Vanning Lead Time": "COMPLETED",
      "TMC Working Day Calendar": "COMPLETED"
    },
    "Grouping Settings": {
      "Grouping": "COMPLETED",
      "Min Max Stock": "IN PROGRESS",
      "Fluctuation Allowance": "NOT STARTED"
    },
    "other": ["Simulation", "Review", "Reports"]
  }
}
```

2. If Monthly NAMC Allocation Plan data is not available:

```
{
  "scenarioId": "uniqueScenarioId",
  "data": [],
  "scenarioSteps": {
    "Line Level Input": {
      "NAMC Production Calendar": "COMPLETED",
      "Model Change Dates": "COMPLETED",
      "NAMC Allocation Plan": "COMPLETED"
    },
    "Vanning Center Inputs": {
      "Shipping Pattern": "COMPLETED",
      "Vanning Lead Time": "COMPLETED",
      "TMC Working Day Calendar": "COMPLETED"
    },
    "Grouping Settings": {
      "Grouping": "COMPLETED",
      "Min Max Stock": "IN PROGRESS",
      "Fluctuation Allowance": "NOT STARTED"
    },
    "other": [
      "Simulation",
      "Review",
      "Reports"
    ]
  }
}
```

viii. If there is an error while fetching the data, an “Internal Server Error” with a status code of 500(*refer: error response-**any internal server error occurred with status 500***) will be returned. Otherwise, a successful response with a status code of 200 will be sent.



SP Monthly NAMC Allocation Plan Get API Techncial workflow

Note:

We will be utilizing this API for both to fetch file data and previous rundown data .

Queries: